

Mohit Appari

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WORK EXPERIENCE

Agency for Persons with Disabilities, State of Florida

Tallahassee, FL

Data Scientist

Jun 2025 - Present

- Build **regression, ARIMA, and vector autoregression (VAR)** forecasting models over **\$200M+ in monthly transaction volume**, projecting service utilization and spend to inform annual budget submissions, provider rate decisions, and capacity planning for a **60,000+ member** population.
- Architect **40+ automated ETL pipelines** in Python, SQL Server, and Azure Data Warehouse processing **300K+ monthly records**, converting a legacy master data management system and manual report generation into scheduled, monitored workflows.
- Re-engineer the agency's heaviest SQL workloads through indexing, partitioning, stored procedures, and execution-plan tuning, **83% runtime reduction, 60 minutes to under 10**, turning an overnight batch process into same-day analysis.
- Lead Power BI development end to end, defining metrics and semantic models behind executive dashboards tracking **60,000+ client applications** and **\$180M in emergency financial assistance**, replacing manual pulls with self-serve visibility for program directors.
- Investigate anomalies across a **60,000+ client dataset**, isolating utilization outliers and data inconsistencies, then correcting them at the source to raise accuracy of agency-wide reporting.
- Translate functional and non-functional requirements from business analysts, project managers, and external stakeholders into scoped analyses and dashboards, applying **CI/CD practices** to analytics deployment.

Florida State University

Tallahassee, FL

Machine Learning Research Assistant

Jun 2024 - Sep 2024

- Fine-tuned deep learning interatomic potentials (**NequIP**) via transfer learning on university HPC clusters across an **11,000+ material dataset**, improving prediction consistency **15%** through systematic hyperparameter optimization.
- **Researched** and benchmarked **generative architectures (GANs, VAEs, RNNs)** to synthesize high-dimensional molecular data, evaluating architecture tradeoffs against predictive performance for downstream molecular dynamics work.
- Designed controlled experiments across model architectures using **statistical hypothesis testing and A/B comparison**, tracked in **MLflow**, with out-of-sample validation protocols.

S&P Global

Bangalore, IN

Software Developer

Jun 2022 - Jul 2023

- Built and shipped a production data capture platform processing **10,000+ financial filings and business documents per day**, using **OCR and CNN-based classification** to convert unstructured and scanned filings into structured data feeding S&P Global's downstream research and analytics products.
- Designed a modular Python extraction framework with **CDC-based incremental updates and 80% test coverage**, cutting the engineering cost of onboarding new document types across multiple business domains.
- Engineered cloud data pipelines on **AWS (S3, Athena) and BigQuery** to ingest, transform, and serve large-scale datasets, enabling near real-time availability for downstream financial data consumers.
- Developed **REST API integrations with schema validation and GitHub Actions CI/CD**, improving data quality and deployment reliability for consuming analytics teams.
- Led a **team of 5 engineers** through the legacy .NET-to-Python migration, standardizing the extraction stack and training the team on the new tooling.

Software Developer Intern

Jan 2022 - Jun 2022

- Researched open-source Python tooling to replace the team's legacy .NET extraction architecture, prototyping the document and image capture approach that became the production platform; **converted to full-time on the same team**.

LG Electronics

Bangalore, IN

Software Developer Intern

May 2021 - Aug 2021

- Built a Python computer vision and IoT prototype for logistics monitoring using **YOLO, REST APIs, PostgreSQL, and Raspberry Pi**, implementing real-time asset tracking and driver drowsiness detection at **90% accuracy**; placed **3rd in the LG Campus Ideathon**.

ITC Limited

Bangalore, IN

Software Developer Intern

Jan 2021 - Mar 2021

- Built a **master data management application** using Flask, MySQL, and the MERN stack with REST APIs and role-based authentication, plus an admin dashboard for permissions, order tracking, and workflow automation that eliminated manual handoffs and the errors they introduced.

EDUCATION

Florida State University, Masters in Data Science, Tallahassee, FL

Aug 2023 - May 2025

Coursework: Stochastic & Markov Processes, Bayesian Inference, Time Series Analysis, Statistical Inference, Machine Learning, Business & Finance

Christ University, Bachelors in Computer Science, Bangalore, IN

Aug 2018 - May 2022

Certification: Akuna Capital Options 101

PROJECTS

RAG-Based LLM ChatBot System | Python, LangChain, OpenAI GPT-4, Cohere, Pinecone, HuggingFace, Streamlit

Built and deployed an end-to-end retrieval-augmented generation system for semantic search and context-aware question answering over unstructured course materials, using Cohere embeddings and Pinecone vector search to serve a class of 40+ students.

Additional projects and portfolio: moh1tt.xyz

SKILLS

Python, SQL, R, PySpark, PyTorch, TensorFlow, scikit-learn, XGBoost, MLflow, LangChain, ARIMA, VAR, GARCH, Claude, OpenAI, TypeScript, FastAPI, REST APIs, Spark, Kafka, Airflow, dbt, Snowflake, PostgreSQL, MongoDB, Redis, Databricks, Docker, Kubernetes, Git, Microservices